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MILITARY SPECIFICATION

DELAY ELEMENT, CARTRIDGE, M5 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M5 Delay Element (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-B-550	Barium Chromate
MIL-A-2550	Ammunition, General Specification for
MIL-C-13739	Composition, Delay
MIL-B-51092	Boron, Amorphous, Powder
MIL-C-60539	Coating, Anodic, Conventional for Aluminum and Aluminum Alloy

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-453	Inspection, Radiographic
MIL-STD-810	Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1377

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2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL ORDNANCE STATION (CAGE Code 19200 and 19203)

DL8594080	Delay Element, Cartridge, M5, Assembly
8826728	Ammunition, Container, Metal, M21 for Delay Element, M5 or M8
8826729	Box, Packing, Ammunition, for Delay Element, M5 or M8 in Metal Container M21

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Data Control Branch (Code 3720), Indian Head, MD 20640-5000.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a first article sample shall be subjected to first article inspection (see 4.4, 6.2, 6.3, and 6.7).

3.2 Conformance to documents. The M5 delay element covered by this specification shall be manufactured in accordance with DL8594080 and all documents listed thereon (see 6.8 through 6.11).

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Components.

3.4.1 Primary components. For the purpose of this specification, the primer, delay charge and the output material charge are considered primary components. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

3.4.1.1 Primer. The primer shall conform to the applicable drawing and reference specification. Only primers from lots manufactured within the last 2 years, or recertified within 1 year, shall be used in these units. Prior to assembly into a delay element, the primer shall be visually examined in accordance with 4.6.5 and radiographically examined for defects as specified in 4.7.1.

3.4.2 Protective finish. The finish of each component part shall be in accordance with MIL-C-60539 and the applicable drawing (see 4.7.2).

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3.5 Assembly.

3.5.1 Radiographic examination. Each delay element assembly shall be radiographically examined for defects (see 4.7.3).

3.5.2 Vibration. The assembled cartridge shall withstand vibration as prescribed in MIL-STD-810, Method 514.3, Test Procedure I, Test Condition I—3.2.5 without evidence of any disintegration of the primer, propellant, or igniter, or derangement of any component (see 4.7.4 and 6.2).

3.6 Functional requirements.

3.6.1 Delay element performance.

3.6.1.1 Misfires. The delay element shall not fail to fire.

3.6.1.2 Ballistics. The delay charge shall consist of barium chromate, as specified in MIL-B-550, and boron, as specified in MIL-B-51092, in the proportion and quantity fixed by the contractor to produce the delay times (see 6.6.1) required in table I (see 6.8 and 6.9).

3.6.1.3 Ballistics (alternative). Delay charge shall consist of barium chromate, potassium perchlorate, and zirconium-nickel-alloy Type I and Type II, as specified in MIL-C-13739, in the proportion and quantity fixed by the contractor to produce the delay times required in table I (see 6.10 and 6.11).

TABLE I. *Ballistic requirements.*

Temperature (°F)	Delay times
−60 to −70	2.1 ± 0.3
65 to 75	2.0 ± 0.2
195 to 205	1.8 ± 0.4

3.7 Workmanship. Delay elements shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.7.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.7.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550 and MIL-STD-453.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified in the contract or purchase order (see 6.2), inspection conditions shall be as follows:

a. Temperature: Room ambient 20 to 25 °C (68 to 77 °F)
b. Altitude: Sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric: 14.7 psia

4.4 First article inspection. At the beginning of regular production, the supplier shall submit a first article sample to a Government approved facility for evaluation. The sample shall consist of 23 delay bodies, 23 primer retainers to be tested as indicated in table II, and 10 sets of unassembled components (including explosive components) for visual and dimensional examination. The first article sample shall be manufactured using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. When it becomes necessary to make a change in either the primer, propellant, or igniter lot after initial production, 12 assembled delay elements from the first production lot thereafter shall be tested in accordance with 4.7.5. The Government reserves the right to require new first article samples until such time as an acceptable sample is submitted (see 6.3).

TABLE II. *First article inspection.*

Paragraph		Test	Sample size	Type of test	
Requirement	Test			Destructive	Nondestructive
3.4.2	4.7.2	Protective finish	23 Delay bodies	X	—
			23 Primer retainers	X	—
3.5.2	4.7.4	Vibration	12 Fr. Ballistics sample	—	X
3.6	4.7.5	Function tests	30	X	—

4.4.1 Examination, tests, and facility. After provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or other such facility specified in the contract (see 6.2).

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4.4.2 Temperature conditioning for ballistic test. The sample shall be temperature conditioned and tested to determine compliance with 3.6.1.2. The sample for firing shall be conditioned to table III.

TABLE III. *Temperature conditioning.*

Temperature (°F)	Not vibrated	Vibrated
−60 to −70	6	4
65 to 75	6	4
195 to 205	6	4

4.4.3 First article failure. Failure of the sample to comply with the requirements of the drawings and specifications shall result in first article sample disapproval.

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. The product shall be submitted in accordance with MIL-STD-105.

4.5.1.2 Lot size. The production lot size shall be as specified in the contract (see 6.2).

4.5.2 Examination and tests. The quality conformance sample shall be subjected to the examinations of 4.6 and all the test procedures of 4.7 (with the exception of 4.7.4).

4.6 Examination. All delay elements and components shall be examined visually and radiographically and characteristics defined in accordance with the applicable drawings. One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4×) magnification may be used in performing visual examination for such defects as burrs and foreign matter. When compliance is in doubt as a result of visual inspection, the characteristic shall be measured or gaged to determine acceptability. All nonconforming items shall be rejected.

4.6.1 Delay Element, M5, Assembly Drawing 8594080.

Characteristic	Defect	Method of Inspection
Critical		
1	Any part missing	X-ray/Visual
2	Improper assembly	X-ray/Visual
3	Primer not properly seated	X-ray/Visual
4	Mixed ammunition	Visual
5	Flash hole missing	Visual
6	Primer uncrimped or seal missing	Visual
7	Marking missing incorrect or illegible	Visual
8	Delay charge not visible through flash hole	Visual

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Characteristic	Defect	Method of Inspection
Major		
101	Concentricity of head with body	Gage
102	Diameter of body, max	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter present	Visual
Minor		
201	Total length, max	Gage
202	Laquer missing or inadequate	Visual

4.6.2 Body Delay, Drawing 8594081, covering a detail of Drawing 8594080.

Critical		
1	Flash hole missing or obstructed	Visual
2	Metal defective	Visual
Major		
101	Pitch diameter, external thread, min	Gage
102	Major diameter, external thread, min	Gage
103	Concentricity of body with pitch diameter of thread	Gage
104	Diameter of cavity, max	Gage
105	Diameter of head, min	Gage
106	Diameter of flash hole	Gage
107	Diameter of thread relief	Gage
108	Length from cavity face to shoulder of head diameter	Gage
109	Protective finish inadequate or defective	Visual
110	Thread damaged or not full	Visual
111	Burr	Visual
112	Foreign matter present	Visual
Minor		
201	Pitch diameter of external thread, max	Gage
202	Diameter of wrench hole	Gage
203	Diameter of thread pilot	Gage
204	Diameter of cavity, min	Gage
205	Diameter of head, max	Gage
206	Diameter of step rear of head	Gage
207	Depth of wrench hole	Gage
208	Depth to base of cavity	Gage
209	Total length	Gage
210	Length of step rear of head	Gage

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Characteristic	Defect	Method of Inspection
211	Length of thread pilot	Gage
212	Width of thread relief	Gage
213	Location of wrench hole	Gage
214	Radius or chamfer missing or incorrect	Visual
215	Surface finish improper	Visual

4.6.3 Gasket, Aluminum, Drawing 8594082, covering a detail of Drawing 8594080.

Critical

None defined

Major

101	Diameter, outside, min	Gage
102	Diameter, inside, max	Gage
103	Metal defective	Visual
104	Burr	Visual
105	Foreign matter present	Visual

Minor

201	Diameter, outside, max	Gage
202	Diameter, inside, min	Gage
203	Thickness	Gage
204	Surface finish improper	Visual

4.6.4 Primer, Retainer, Drawing 8594083, covering a detail of Drawing 8594080.

Critical

1	Flash hole missing or obstructed	Visual
2	Depth of primer pocket	Gage
3	Metal defective	Visual

Major

101	Pitch diameter of internal thread, max	Gage
102	Minor diameter of internal thread, max	Gage
103	Concentricity of head with body	Gage
104	Concentricity of pitch diameter of thread with body	Gage
105	Concentricity of primer pocket with body	Gage
106	Concentricity of recess around primer pocket with primer pocket	Gage
107	Diameter of body, min	Gage
108	Diameter of head	Gage
109	Diameter of primer pocket	Gage
110	Diameter at base of angled recess	Gage
111	Diameter of recess around primer pocket	Gage
112	Depth of cavity	Gage
113	Depth of recess around primer pocket	Gage

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Characteristic	Defect	Method of Inspection
114	Depth of chamfer at primer pocket	Gage
115	Length of effective thread, min	Gage
116	Location of datum diameter	Gage
117	Thread damaged or not full	Visual
118	Diameter of flash hole	Gage
119	Symmetry of flat with body	Gage
120	Protective finish inadequate or defective	Visual
121	Surface finish of primer pocket improper	Visual
122	Burr	Visual
123	Foreign matter present	Visual
Minor		
201	Pitch diameter of internal thread, min	Gage
202	Diameter of body, max	Gage
203	Total length	Gage
204	Length of wrench flats	Gage
205	Width across wrench flats	Gage
206	Location of wrench flats	Gage
207	Radius or chamfer missing or incorrect	Visual
208	Surface finish improper, except as otherwise classified	Visual

4.6.5 Primer Assembly, Percussion, Drawing 5184975, covering a detail of Drawing 8594080.

Critical		
1	Mixed primers	Visual
2	Primer pellet broken, cracked, or dislodged	Visual
3	Primer foil discolored	Visual
4	Metal defective	Visual
Major		
101	Burr	Visual
102	Foreign matter present	Visual
Minor		
None defined		

4.6.6 Ammunition Container, Metal M21 for Delay Element M5 or M8, Drawing 8826728.

Critical	
None defined	
Major	
None defined	

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Minor

201	Contents loose	Manual/Visual
202	Separator or filler missing	Visual
203	Sealing inadequate, incompetent, or incorrect	Visual
204	Container cut or damaged	Visual
205	Marking missing, incorrect, or illegible	Visual

4.6.7 Box, Packing Ammunition for Delay Element M5 or M8 in Metal Container, M21, Drawing 8826729.

Critical

None defined

Major

None defined

Minor

201	Contents loose	Manual/Visual
202	Hardware missing, loose, broken, or improperly engaged	Visual
203	Strapping missing, loose, broken, or improperly assembled	Visual
204	Contents of box exposed	Visual
205	Box damaged	Visual
206	Marking missing, incorrect, or illegible	Visual
207	Number of cartons incorrect	Visual

4.7 Test methods and procedures.

4.7.1 Primer x-ray. Prior to assembly into the cartridge, the primer shall be radiographically examined in accordance with MIL-STD-453. Each primer shall be x-rayed on boards or trays with the cup side perpendicular to the rays. The x-ray negatives shall be identified by item nomenclature and by lot number. Primers are to be left mounted to the boards or trays until after the x-rays have been reviewed and defects identified to facilitate removal of defective units. After screening, the x-ray negatives with defectives identified, shall be forwarded to the engineering activity designated by the acquiring activity (see 6.2). Interpretation of the radiographic films shall verify the following:

Presence of Primer Charge
No Material Defects

4.7.2 Protective finish. Unless otherwise specified (see 6.2), sampling plans and procedures in the determination of acceptability of coated parts shall be in accordance with the provisions of the contract and MIL-C-60539.

4.7.3 Delay element x-ray. Prior to assembly into a cartridge, the delay element shall be radiographically examined in accordance with MIL-STD-453. Each delay element shall be x-rayed with the sides of the delay element perpendicular to the rays. The x-ray negatives shall be identified by item nomenclature and by lot number. Delay elements are to be left mounted to the boards or trays until after the x-rays have been reviewed and defects identified to facilitate removal of defective units. After screening, the x-ray negatives with defectives identified, shall be forwarded to the engineering activity designated by the acquiring activity.

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4.7.4 Vibration test. Twelve of the 30 assembled delay elements of the first article sample, x-rayed as directed in 4.7.3, properly identified, shall be subjected to vibration. The method and procedures for performing the vibration test shall be in accordance with MIL-STD-810, Method 514.3, Test Procedure I, Test Condition I-3.2.5 (see 6.2). Then the 12 vibrated delay elements shall be x-rayed again. Evidence of any disintegration of the primer, propellant, or derangement of any component shall be cause for rejection of the sample.

4.7.5 Functional. The following tests shall be performed by the Government.

4.7.5.1 Ballistics

4.7.5.1.1 First article. Thirty assembled delay elements of the first article sample shall be grouped and temperature conditioned as described in table III and 4.8. Following removal from the conditioning chamber, the cartridge shall be tested as required by 3.6.1.2. Failure of delay element to comply with the requirements of 3.6.1.2 shall be cause for sample rejection.

4.7.5.1.2 Production lot acceptance. Thirty assembled delay elements from each production lot shall be grouped and temperature conditioned in accordance with 3.6.1.2. Failure of a delay element to comply with 3.6.1.2 shall be cause for rejection of the lot. In addition to the sample of cartridges for the above test, 10 additional cartridges shall be furnished from each prodution lot for investigative purposes.

4.7.5.2 Retest. There shall be no retests.

4.7.5.2.1 Test failure. If a test failure is attributable to an assignable cause, excluding the test cartridge, the original test results shall be discarded and that test reconducted.

4.8 Temperature conditioning. The conditioning chamber shall be preconditioned to the specified temperature. Once the temperature inside the chamber has stabilized, the samples shall be placed inside and maintained at the specified temperature for 4 hours minimum, 24 hours maximum.

4.8.1 Reconditioning. The sample must be tested within five minutes after removal from the conditioning chamber. If this time is exceeded, the cartridge shall be reconditioned in accordance with 4.8.

4.9 Sampling and visual inspection. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be in accordance with Drawings 8826728 and 8826729, except that the marking shall refer to delay element, cartridge, M5 assembly.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The assembly is used in cartridge actuated devices for Aircrew Emergency Escape Systems.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and revision letter of this specification
b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced and that MIL-STD-810D shall be used (see 2.1, 3.5.2, and 4.7.4)
c. Whether a first article sample is required (see 3.1, 4.4, and 6.3)
d. Test facility for first article tests (see 4.4.1)
e. Production test activity and lot size (see 4.1 and 4.5.1.2)
f. Any special test conditions (see 4.3)
g. Directions for shipping radiographs of primers and of entire lot of delay element assemblies (see 4.7.1 and 4.7.3)
h. Directions for sampling for protective finish tests (see 4.7.2)
i. Level of preservation and packing (see 5.1)
j. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition and Explosives" (DOD 4145.26M) are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5 are applicable.

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Explosive hazard classification information.

Hazard Classification Division: 1.4
Storage Compatibility Group: S
DOT Class: C
DOT Marking — Explosive Power Device — Class C — Handle Carefully — Keep Fire Away

6.5 Hazard notice. The cartridge described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.6 Definition.

6.6.1 Delay time. Delay time is defined as the time from primer impact to the start of pressure rise.

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6.7 Manufacture as a component. When the delay element is being manufactured in conjunction with a delay cartridge as a component thereof, the first article requirements for ballistic, vibration, x-ray, and packaging requirements are waived, and the requirements of the cartridge specification shall apply.

6.8 Advisory output material (formulation). The advisory output material (formulation) must be approximately 23.7 percent boron, 70.7 percent potassium nitrate, and 5.6 percent binder. Advisory individual delay element output material weight must be approximately 1.5 grains.

6.9 Advisory delay charge composition (formulation). The advisory delay charge composition must be approximately 97.0 percent barium chromate and 3.0 percent boron. Advisory individual delay element charge weight must be approximately 18 to 20 grains.

6.10 Advisory output material. The advisory output material must be three pellets in accordance with Drawing 11738756.

6.11 Advisory individual delay charge composition (formulation). The advisory individual delay element charge weight must be approximately 6 grains Type I, 10 grains Type II, and 6 grains Type I comprising constituents specified in MIL-C-13739.

6.12 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.13 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:
Navy-OS
Air Force-99

Review activity:
Air Force-70

Preparing activity
Navy-OS
(Project 1377-0979)

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